Final Design Project Report

EEE 202 Lab (9:00-12:00 pm Tue)

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Introduction:

The purpose of this lab and design project was to design, build, and test an automated solar power meter that measured voltage, current, and power in real time. This design was intended to streamline the use of solar panels, allowing for better usability when compared to traditional methods of having to calculate power current and voltage values by hand. By the integration of a microcontroller with analog circuits, a more user-friendly solution was developed.

This project took into account multiple concepts that we have learned throughout the lecture component of the course. Which included voltage dividers, operational amplifiers, and first-order transient circuits. Furthermore, the project required knowledge of circuit construction, debugging code, and comparing theoretical results with real-world measurements. The final prototype made use of an Arduino Uno R3 to process the analog signals and display the results on the LCD screen as well as the serial monitor.

Design Intent and Stakeholders:

The primary focus of the design was to create a portable and efficient way of calculating the voltage current and power that was generated by a solar panel. The intended users for this project were solar panel installers or anyone who needed an effective and efficient way of measuring important data from a solar panel on the spot.

The most important considerations for the design of the solar panel meter were the accuracy, simplicity, and portability of the design. The measurements had to be close to the theoretical results, and the system had to work automatically in real time suitable for field testing.

If the design were to be scaled up for a larger solar panel system as opposed to just one, additional factors would have to be considered. The most important ones being the handling of higher voltages and currents and the cost effectiveness of the design when it comes to mass production. Safety would also have to be a concern, as we would be dealing with higher power levels.

Technical Design and Theory:

The overall design consisted of different components, with the most important ones being a voltage divider for the voltage measurement, a shunt resistor and an operational amplifier for current measurement, an Arduino for data processing, and an LCD display for displaying the readings.

Voltage Divider Theory/Logic:

A voltage divider is a circuit that is used to scale down a voltage using two resistors that are in series with one another. The output voltage can be given by the following equation:

$$V = Vi \cdot R2\frac{1}{R1+R2}$$

This relationship shows that the voltage depends on a ratio of resistors and not their absolute values.

In this project the solar panel was capable of producing a max voltage of 6.9V under full sunlight, but the Arduino pins can only take values that are up to 5V. The voltage divider was therefore used to reduce the input voltage. To achieve this, resistor values of 2.2K and 4.7K were selected.

Current Measurement and Op Amp Theory:

To measure current in our design, we employed the use of a shunt resistor that was placed in series with the load. In accordance with Ohm's law, V = IR, the voltage across the shunt resistor is proportional to the current that flows through it.

However, we encounter a problem as the shunt resistor has a small resistance value of 47 ohms; because of this, the voltage would also be very small, something that an Arduino has difficulty measuring. This is where the op-amp comes in, as it allows us to bypass this issue by amplifying the signal.

The non-inverting op-amp was chosen, where the gain is given by the following:

$$G = 1 + \frac{Rf}{Ri}$$

The resistor values that were selected would produce a gain of roughly 5.5. Which achieved the goal of increasing the small shunt voltage into something that could be accurately measured by the Arduino microcontroller without exceeding its 5V limit.

Note that the op-amp used was non-ideal, meaning that the input current is around zero and the voltage difference between the inputs is also zero. By using the non-ideal op-amp, the design process was simplified.

Arduino Data and Signal Processing:

In order to process the data that the Arduino would be receiving, the microcontroller would be using an analog-to-digital converter, which converts input voltages up to 5V into digital values that range up to 1023. The conversion used is given by:

$$V = ADCreading \cdot \frac{5}{1024}$$

The Arduino code would then revert the effects of the voltage divider and op-amp gain in order to calculate the actual solar panel voltage and current.

For power calculations the following equation was used:

$$P = VI$$

By doing these calculations, the microcontroller was able to successfully display the current voltage and power automatically on the LCD display.

LCD display design integration:

The LCD was used as a means to display the calculated data that the microcontroller would output. It presented the necessary values in real time all while not requiring the Arduino to be constantly connected to a computer.

The Display Read:

- Voltage (V)
- Current (mA)
- Power (mW)

The LCD usage was an easy way to display the necessary data in a clear and practical way.

Transient Analysis and Capacitor Selection:

A capacitor was included in the circuit design; its function was to control the discharge behavior when the load was disconnected. When discharging a capacitor, the voltage across it can be given by:

$$V(t) = Voe^{-t/RC}$$

In order to meet the design constraints, the capacitor needed to decay close to zero in this case, about 1 mV, within 10 ms. Using the time constant:

$$\tau = RC$$

We were able to select a capacitor value that met the design requirements, which turned out to be approximately $1\mu F$ allowing our system to work as intended.

Engineering Decisions and Design Trade-Offs:

During the design process there were some noteworthy trade-offs. Perhaps the largest one was the selection of the resistors. While ideal resistor values could be easily found and calculated, the design had to take into consideration the materials that were available in the lab. As a result, the design was adjusted accordingly while meeting the design criteria.

Op amp gain was another area where a trade-off was involved more specifically concerning the op amp gain. While a higher gain would improve the measurements, we had to keep in mind that the Arduino had a voltage limit of 5V. Thus, a gain of about 5.5 was chosen as a middle ground.

The shunt resistor also had its setbacks; the larger the resistor, the higher the voltage signal, wasting more power. Smaller resistors minimize this loss but are in need of amplification.

Verification and Validation of Experimental Results:

During testing different lighting conditions were considered.

The measured values were the following:

- Full Sunlight: 6.98 V and about 7.1 mA
- Shaded environments: 3.1 V and 5.8 mA

For example:

- Simulated voltage was about 7.33 V
- The Measured voltage would be slightly less 7.27 V

These discrepancies are to be expected, as the condition of components was not an ideal scenario; for instance, our resistors were not exactly 220 ohms but slightly less, around 217 ohms. Another factor is varying light conditions different from the constant conditions in the simulations. These small discrepancies led to small errors, which were to be expected.

Troubleshooting and design iterations:

The issues encountered during the testing phase involved an issue with the LCD display not functioning properly and displaying the data. This was quickly fixed by swapping it out for another display, which resolved all issues.

The different components used in Tinkercad vs. the laboratory components that were available were another issue, as the Tinkercad operational amplifier is different from the physical one in the lab. This was an oversight in the design, as it was assumed to be the same component. This was later fixed by reconnecting wires in their proper spots, allowing the circuit to properly display the current and power, as it previously read 0 for both.

Professional Growth:

This project was an opportunity to apply the engineering skills and techniques that I have learned before and expand on them into the realm of circuit design and prototyping with iterations.

My debugging skills were expanded when it came to helping with the code required for the microcontroller to display the correct results. Identifying small issues in the code was key to a successful design implementation. Moreover, the project stressed the comparison between simulated results and actual results that were collected from our physical build. This project proved that while simulations are an important step in the design process, the physical aspect is just as important as identifying points of differences and allows for better design optimization.

The experience using a common microcontroller was also welcomed, as it allows for the design of future projects to be a more welcome experience rather than unknown territory. Overall the project was a great way to learn and apply the skills learned in class and the laboratory.

Conclusion:

At the end of the project, the design for the solar panel power meter was successfully implemented. The system successfully read the correct and expected voltage, power, and current readings in real time. Although there were discrepancies, they were to be expected given the constraints of component selection and quality. The project successfully demonstrated a practical application of circuit theory. The lab was effective in connecting in-class theory to real-life engineering problems.